

KAUSHAL BHANDERI

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EDUCATION

Charotar University of Science and Technology (CHARUSAT)

August 2022 - April 2026

Bachelor in Computer Engineering (CGPA: 9.81 / 10.00)

EXPERIENCE

Research Intern @Speech and Information Processing Lab, IIT Hyderabad

February 2026 - June 2026

Soundfile, GURadio, Scipy, PyTorch

Automatic Speech Recognition

- Created a 100+ hour synthetic military speech dataset by modeling radio communication artifacts such as channel noise, channel modulation, compression effects, distortion, packet loss, and bandwidth degradation.
- Optimized Whisper-Small for military speech recognition, achieving 0.23 WER, and validated performance on a manually transcribed 9-hour real walkie-talkie speech benchmark curated from publicly available communication recordings.

AI Engineer Intern @Gurugrace Consultancy Service - Remote

December 2025 - June 2026

PyTorch, Transformers, Qwen, Mistral

Emotion-Aware AI

- Architected an Emotionally Aligned AI framework for LLMs using activation steering, prompt engineering, and sparse emotion circuit interventions, achieving 97.4% emotion-expression accuracy on Mistral-3-3B and 99.4% on Qwen3-4B-Instruct.
- Built emotion-aware conversational AI systems through a RoBERTa-based emotion classifier (98.8% F1), target classifier (96.6% F1), LoRA-fine-tuned SenseVoice for Speech Emotion Recognition, and controllable emotion-aware TTS using CosyVoice2 and F5-TTS.

Research Intern @LTRC, IIIT Hyderabad

May 2025 - December 2025

Fairseq, Soundfile, PyTorch, Transformers

Speech-to-Speech Translation

- Conducted extensive evaluation of direct and cascade speech-to-speech translation architectures across Indic-Indic, Indic-English, and English-Indic language pairs, achieving competitive performance and surpassing AI4Bharat baselines on selected BhashaAnuvad benchmarks.
- Fine-tuned Seamless models and standardized large-scale multilingual speech datasets for direct Indic-English speech-to-text translation.

Data Science Intern @Cloud Express Solution - Remote

September 2024 - December 2024

Stable-Baselines3, Gymnasium, Flask

Reinforcement Learning

- Processed 2000+ rweb interaction logs to build reinforcement learning-based automated website versioning systems driven by user behavior analytics.
- Deployed a Flask API to deploy RL-generated website updates every 2 hours, reducing downtime by 30% and improving deployment efficiency.

PROJECTS

PrepArc AI-Based Interview System | NLP, LLM, Computer Vision, Speech Analysis, Resume Parsing

- Formulated an AI-driven interview platform with automated question generation, resume parsing, and ATS score prediction using LLMs.
- Implemented post-interview analysis using facial expression recognition and speech analytics, achieving 70% confidence and tone detection accuracy.

HealAI Personalized Medical Chatbot | NLP, LLM, Speech-to-Text, Text-to-Speech, Multilingual Support

- Engineered a personalized multilingual medical chatbot with voice interaction capabilities, supporting 5+ indian languages, enabling seamless real-time medical consultations and enhancing accessibility for 100+ users.
- Integrated advanced NLP models and LLMs to accurately interpret medical queries with 93% accuracy, providing contextually relevant responses and improving user engagement by 80%.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, Python, SQL

Technologies: ROS2, Tensorflow, Keras, NLTK, Pytorch, Transformers, Spacy, OpenCV, Github

Concepts: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Speech Processing, Computer Vision, Generative AI, Agentic AI

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [C.1] **Kaushal Bhanderi, Ronak R. Patel, Arpita Shah (2025). Attention-Enhanced UNETR for Polyp Segmentation: Improving Accuracy and Robustness in Medical Imaging.** In *5th International Conference on Computer Vision and Robotics (CVR 2025)*, National Institute of Technology Goa.